

A Cryptographic Infrastructure for Friend-to-Friend Choreographies

Vincenzo Ciancia@ISTI - CNR

Roberto Guanciale@KTH

Emilio Tuosto @GSSI

LAP 2026

23-27 September 2026, Dubrovnik, Croatia

A framework for

A framework for
friend-to-friend

A framework for
friend-to-friend
local-first

A framework for

friend-to-friend

local-first

coordination

A framework for

friend-to-friend

local-first

coordination

and its (forthcoming) implementation

*To appear on
JLAMP 2026*

- ▶ Motivations to go friend-to-friend
(and not being "obscured by clouds")

*To appear on
JLAMP 2026*

- ▶ Motivations to go friend-to-friend
(and not being "obscured by clouds")
- ▶ A model for local-first coordination...
 - ▶ F2F-choreographies
 - ▶ Nearbybytes

*To appear on
JLAMP 2026*

- ▶ Motivations to go friend-to-friend
(and not being "obscured by clouds")
- ▶ A model for local-first coordination...
 - ▶ F2F-choreographies
 - ▶ Nearbybytes
- ▶ Cryptographic scaffolding

*To appear on
JLAMP 2026*

Blessing and curse of clouds

Advantages of clouds

cloud computing = fast communication infrastructures + global-scale computation facilities

Advantages of clouds

cloud computing = fast communication infrastructures + global-scale computation facilities

- ▶ software-as-service

Advantages of clouds

cloud computing = fast communication infrastructures + global-scale computation facilities

- ▶ software-as-service
- ▶ dynamic reconfiguration (services can be replaced if needed)

Advantages of clouds

cloud computing = fast communication infrastructures + global-scale computation facilities

- ▶ software-as-service
- ▶ dynamic reconfiguration (services can be replaced if needed)
- ▶ reduced costs for maintenance, facilities, personnel, software, and hardware

Advantages of clouds

cloud computing = fast communication infrastructures + global-scale computation facilities

- ▶ software-as-service
- ▶ dynamic reconfiguration (services can be replaced if needed)
- ▶ reduced costs for maintenance, facilities, personnel, software, and hardware
- ▶ anytime-anywhere motto

No free lunch

No free lunch

- ▶ Lack of transparency

Cloud-computing is inherently a "black-box"

No free lunch

- ▶ Lack of transparency

Cloud-computing is inherently a "black-box"

- ▶ Vendor lock-in

Switching to a different vendor can have prohibitive costs

No free lunch

- ▶ Lack of transparency

Cloud-computing is inherently a "black-box"

- ▶ Vendor lock-in

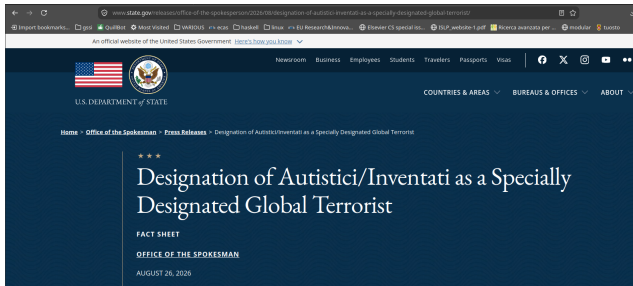
Switching to a different vendor can have prohibitive costs

- ▶ Dependence on connectivity

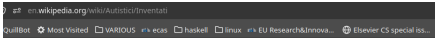
Cloud-based computing is heavily reliant on network connectivity

...anytime-anywhere almost always

...and limited autonomy/sovereignty



...and limited autonomy/sovereignty



Autistici/Inventati

Article Talk

From Wikipedia, the free encyclopedia

"A/I" redirects here. For other uses, see *AI (disambiguation)*.

This article has an unclear citation. The references used may be made clearer by using a different or consistent style of citation and footnoting. (September 2026) [Learn how and how to remove this message](#)

Autistici/Inventati (A/I) was an Italian hacktivist group that provided Internet communication services for groups opposed to fascism, militarism, racism, sexism, homophobia and transphobia.^[1]

Autistici/Inventati	
Abbreviation	A/I
Established	2010
Dissolved	6 September 2016
Type	Non-profit
Services	Encryption, secure communication

...and limited autonomy/sovereignty



Perhaps some re-thinking is needed?

en.wikipedia.org/wiki/Autistici/Inventati

as Monuments: Photograph a monument, help Wikipedia and win!

more

Autistici/Inventati

[Article](#) [Talk](#)

From Wikipedia, the free encyclopedia

"A/I" redirects here. For other uses, see [AI \(disambiguation\)](#).

This article **has an unclear citation**. The references used may be made clearer by using a different or consistent style of citation and footnoting. (September 2026) ([Learn how and how to remove this message](#))

Autistici/Inventati (A/I) was an Italian **hacktivist** group that provided Internet communication services for groups opposed to **fascism**, **militarism**, **racism**, **sexism**, **homophobia** and **transphobia**.^[1]

Abbreviation	A/I
Established	2010
Dissolved	6 September 2010
Type	Non-profit
Services	Encryption, web hosting, and other services

What if we...

What if we...

- ▶ kept reliance on the cloud to a minimum?

What if we...

- ▶ kept reliance on the cloud to a minimum?
- ▶ retained control over
 - ▶ data (i.e., ensure **confidentiality**)
 - ▶ computational resources

What if we...

- ▶ kept reliance on the cloud to a minimum?
- ▶ retained control over
 - ▶ data (i.e., ensure confidentiality)
 - ▶ computational resources
- ▶ had more "autonomy"

Local-first coordination
via
Friend-to-Friend Choreography

Our choreographic model

Syntax

$$\begin{aligned} F &::= A \rightsquigarrow B : t \\ &| F + F' \\ &| F ; F' \\ &| F \mid F' \\ &| \dots \end{aligned}$$

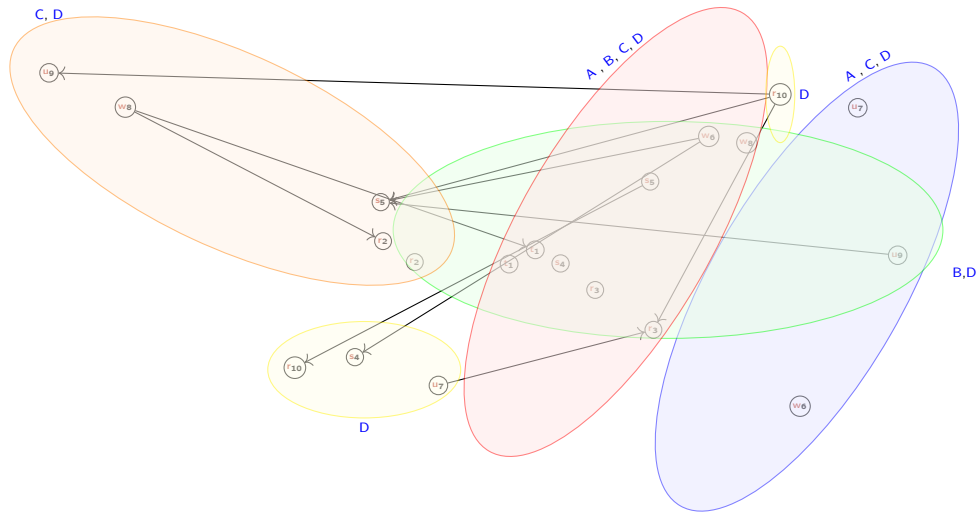
where

- ▶ B is a finite (possibly empty) set of A 's "buddies"
- ▶ t is a "type"

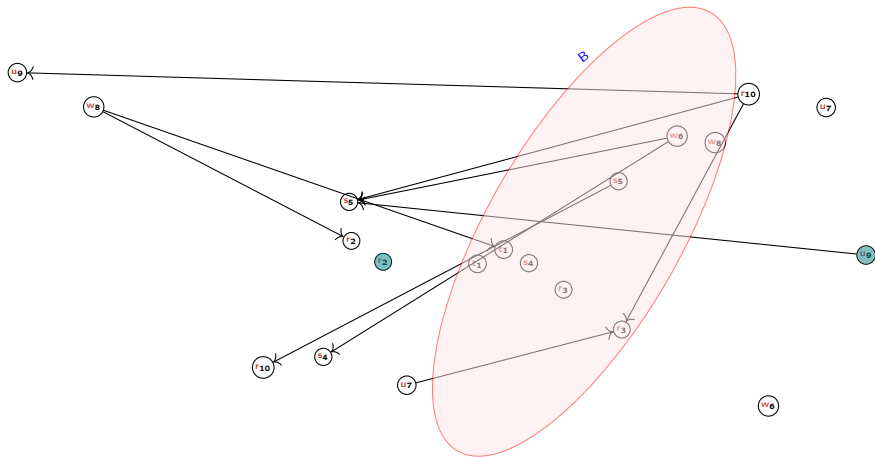
Semantics

$$\begin{aligned} \llbracket A \rightsquigarrow B : t \rrbracket &= \left\{ [A!B\langle t \rangle] \right\} \\ \llbracket F + F' \rrbracket &= \llbracket F \rrbracket \cup \llbracket F' \rrbracket \\ \llbracket F ; F' \rrbracket &= \{ r ;_{\varepsilon_r} r' \mid \dots \} \\ \llbracket F \mid F' \rrbracket &= \{ r ;_{\emptyset} r' \mid \dots \} \\ \llbracket \dots \rrbracket &= \dots \end{aligned}$$

The coordination model, informally



Replication



Having fun together

Having fun together

Alice wants to arrange something for the evening with Bob, Carl, Daphne, and Evelin

Having fun together

Alice wants to arrange something for the evening with Bob, Carl, Daphne, and Evelin

A's options are between going out *for pizza* to a cinema or for a pizza at home

- ▶ if A opts for a cinema, B decides which movie; E should be oblivious of the title
- ▶ otherwise, deal is struck with whoever tells A which pizza they want

*since E would
complain anyway*

A and B share the

pizza

Having fun together

Alice wants to arrange something for the evening with Bob, Carl, Daphne, and Evelin

A's options are between going out to a cinema or for a pizza at home

- ▶ if A opts for a cinema, B decides which movie; E should be oblivious of the title
- ▶ otherwise, deal is struck with whoever tells A which pizza they want

Confidentiality constraints:

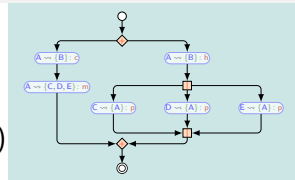
- ▶ E is not involved in the C-branch: can she join?
- ▶ X's choice of pizza is known only to X and A

An F2F-choreography for fun

$$\begin{aligned} F = & \quad A \rightsquigarrow B : c; B \rightsquigarrow \{A, C, D\} : m \\ & + \quad A \rightsquigarrow B : h; (C \rightsquigarrow A : p \mid D \rightsquigarrow A : p \mid E \rightsquigarrow A : p) \end{aligned}$$

An F2F-choreography for fun

$F =$ $A \rightsquigarrow B : c; B \rightsquigarrow \{A, C, D\} : m$
+ $A \rightsquigarrow B : h; (C \rightsquigarrow A : p \mid D \rightsquigarrow A : p \mid E \rightsquigarrow A : p)$



An F2F-choreography for fun

$$F = \begin{array}{l} A \rightsquigarrow B : c; B \rightsquigarrow \{A, C, D\} : m \\ + \quad A \rightsquigarrow B : h; (C \rightsquigarrow A : p \mid D \rightsquigarrow A : p \mid E \rightsquigarrow A : p) \end{array}$$

$$\llbracket F \rrbracket = \left\{ \left[\begin{array}{c} A!B\langle c \rangle \\ \downarrow \\ B!\{A, C, D\}\langle m \rangle \end{array} \right], \left[\begin{array}{c} A!B\langle h \rangle \\ \swarrow \quad \downarrow \quad \searrow \\ C!A\langle p \rangle \quad D!A\langle p \rangle \quad E!A\langle p \rangle \end{array} \right] \right\}$$

An F2F-choreography for fun

$$F = \begin{array}{l} A \rightsquigarrow B : c; B \rightsquigarrow \{A, C, D\} : m \\ + \quad A \rightsquigarrow B : h; (C \rightsquigarrow A : p \mid D \rightsquigarrow A : p \mid E \rightsquigarrow A : p) \end{array}$$

$$\llbracket F \rrbracket = \left\{ \left[\begin{array}{c} A!B\langle c \rangle \\ \downarrow \\ B!\{A, C, D\}\langle m \rangle \end{array} \right], \left[\begin{array}{c} A!B\langle h \rangle \\ \swarrow \quad \downarrow \quad \searrow \\ C!A\langle p \rangle \quad D!A\langle p \rangle \quad E!A\langle p \rangle \end{array} \right] \right\}$$

Let's spell out some requirements:

- ▶ h and c may be public, but let's keep the rest confidential
- ▶ E should be able to know if A decided to go out or B choose the movie
- ▶ C , D , and E cannot choose a pizza until A makes her choice

Nearbytes and its cryptographic infrastructure

What is Nearbybytes?

What is Nearbybytes?

- ▶ set of protocols for confidential interactions combining
 - ▶ communication via "physical wiring"
 - ▶ with cryptography + "hubs" merging

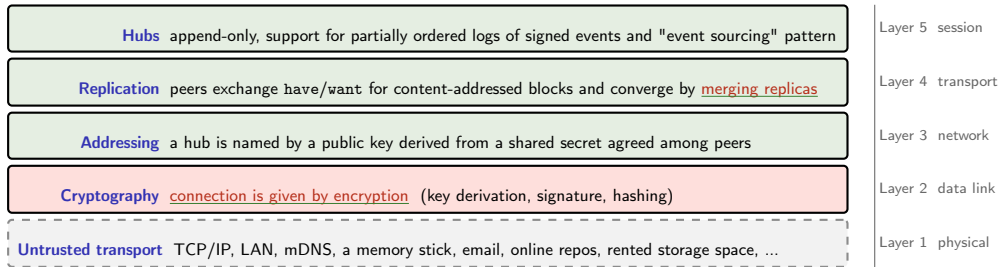
What is Nearbybytes?

- ▶ set of protocols for confidential interactions combining
 - ▶ communication via "physical wiring"
 - ▶ with cryptography + "hubs" merging
- ▶ logical content-based addressing leveraging public-key cryptography

What is Nearbybytes?

- ▶ set of protocols for confidential interactions combining
 - ▶ communication via "physical wiring"
 - ▶ with cryptography + "hubs" merging
- ▶ logical content-based addressing leveraging public-key cryptography
- ▶ data-related events form a partially ordered log

Nearbytes's stack

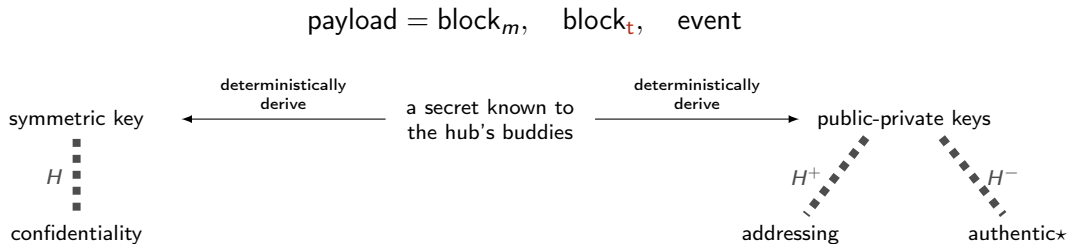


Nearbytes

Hush-hush (no key distribution)

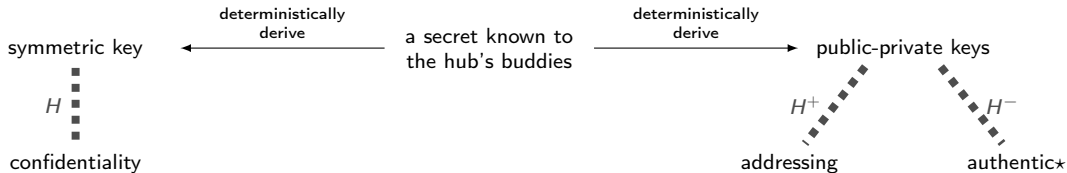
payload = block_m, block_t, event

Hush-hush (no key distribution)



Hush-hush (no key distribution)

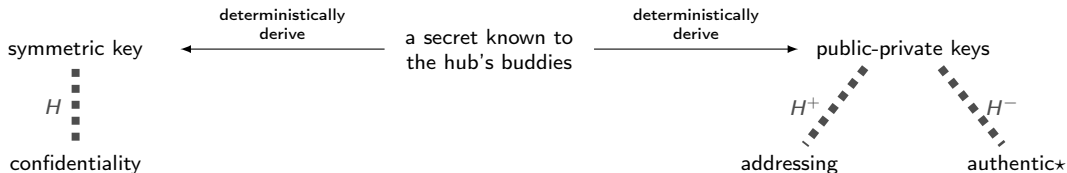
payload = block_m, block_t, event



$$\text{block}_m = \{ \text{msg} \}_H$$

Hush-hush (no key distribution)

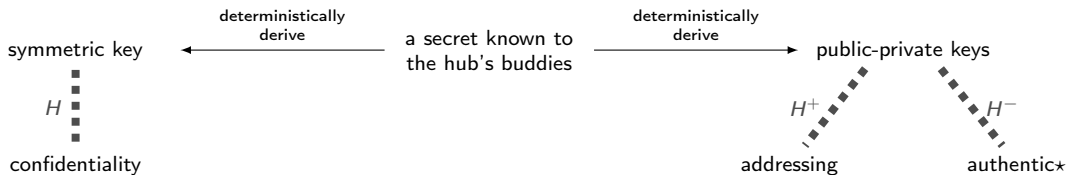
payload = block_m, block_t, event



block_m = { msg }_H
block_t = { type }_{H⁺}

Hush-hush (no key distribution)

payload = block_m, block_t, event



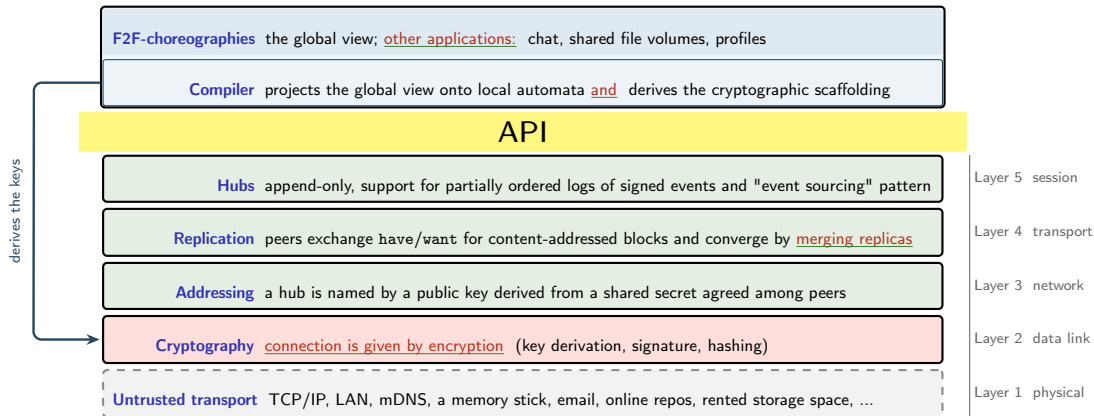
block_m = { msg }_H

block_t = { type }_{H⁺}

event = { hash(ev₁), ..., hash(ev_k), hash(block_m), hash(block_t) }_{H⁻}

F2F-choreographies & Nearbybytes

F2F-choreographies on top of Nearbytes



Hubs' addressing

Buddies agree on a "master secret" M

Leverage buddies' public keys

$$\left\{ \begin{array}{lcl} H & = & M \cdot A^+ \cdot B^+ \cdot C^+ \cdot D^+ \cdot E^+ \\ H_1 & = & M \cdot A^+ \cdot B^+ \cdot C^+ \cdot D^+ \\ \dots & & \\ H_{14} & = & M \cdot A^+ \cdot B^+ \\ H_{15} & = & M \end{array} \right.$$

Projections

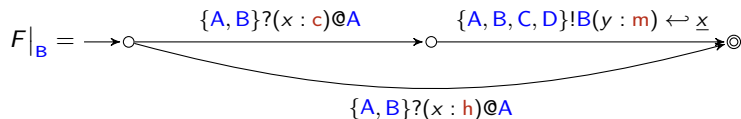
Deriving buddies' expected behaviour:

$$F = \quad A \rightsquigarrow B : c; B \rightsquigarrow \{A, C, D\} : m \\ + \quad A \rightsquigarrow B : h; (C \rightsquigarrow A : p \mid D \rightsquigarrow A : p \mid E \rightsquigarrow A : p)$$

Projections

Deriving buddies' expected behaviour:

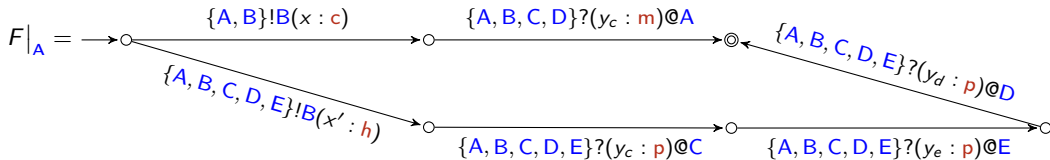
$$\begin{aligned} F = & \quad A \rightsquigarrow B : \textcolor{blue}{c}; B \rightsquigarrow \{A, \textcolor{blue}{C}, \textcolor{blue}{D}\} : \textcolor{brown}{m} \\ & + \quad A \rightsquigarrow B : \textcolor{brown}{h}; (C \rightsquigarrow A : \textcolor{brown}{p} \mid D \rightsquigarrow A : \textcolor{brown}{p} \mid E \rightsquigarrow A : \textcolor{brown}{p}) \end{aligned}$$



Projections

Deriving buddies' expected behaviour:

$$F = \begin{aligned} & A \rightsquigarrow B : \textcolor{blue}{c}; B \rightsquigarrow \{A, C, D\} : \textcolor{brown}{m} \\ & + \quad A \rightsquigarrow B : \textcolor{blue}{h}; (C \rightsquigarrow A : \textcolor{brown}{p} \mid D \rightsquigarrow A : \textcolor{brown}{p} \mid E \rightsquigarrow A : \textcolor{brown}{p}) \end{aligned}$$



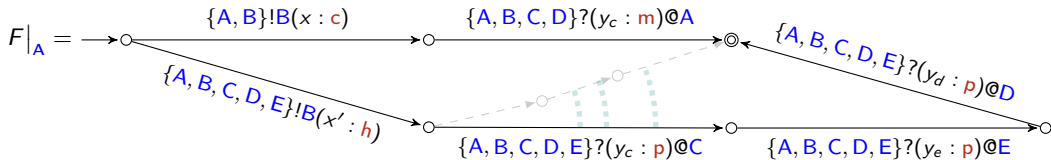
$$\left[A! \{A, B, C, D, E\} \langle \textcolor{blue}{h} \rangle \right]$$

$$e = \{ \textcolor{blue}{A} \text{'s address} \}_H, \{ \textcolor{blue}{h} \}_{H^+}, \{ \text{hash}(\{ \textcolor{blue}{A} \text{'s address} \}_H), \text{hash}(\{ \textcolor{blue}{h} \}_{H^+}) \}_{H^-}$$

Projections

Deriving buddies' expected behaviour:

$$F = \begin{aligned} & A \rightsquigarrow B : \text{c}; B \rightsquigarrow \{A, C, D\} : \text{m} \\ & + \quad A \rightsquigarrow B : \text{h}; (C \rightsquigarrow A : \text{p} \mid D \rightsquigarrow A : \text{p} \mid E \rightsquigarrow A : \text{p}) \end{aligned}$$



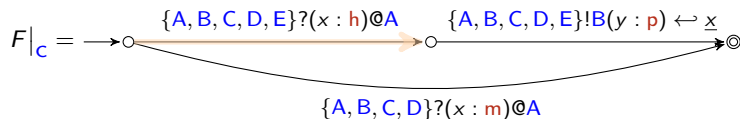
$$\left[A! \{A, B, C, D, E\} \langle \text{h} \rangle \right]$$

$$e = \{A \text{ 's address}\}_H, \{\text{h}\}_{H^+}, \{\text{hash}(\{A \text{ 's address}\}_H), \text{hash}(\{\text{h}\}_{H^+})\}_{H^-}$$

Projections

Deriving buddies' expected behaviour:

$$F = \begin{array}{l} A \rightsquigarrow B : c; B \rightsquigarrow \{A, C, D\} : m \\ + \quad A \rightsquigarrow B : h; (C \rightsquigarrow A : p \mid D \rightsquigarrow A : p \mid E \rightsquigarrow A : p) \end{array}$$



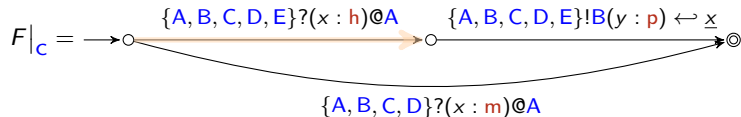
$$\left[A! \{A, B, C, D, E\} \langle h \rangle \right]$$

$$e = \{A \text{ 's address}\}_H, \{h\}_{H^+}, \{\text{hash}(\{A \text{ 's address}\}_H), \text{hash}(\{h\}_{H^+})\}_{H^-}$$

Projections

Deriving buddies' expected behaviour:

$$F = \begin{array}{l} A \rightsquigarrow B : \text{c}; B \rightsquigarrow \{A, C, D\} : \text{m} \\ + \quad A \rightsquigarrow B : \text{h}; (C \rightsquigarrow A : \text{p} \mid D \rightsquigarrow A : \text{p} \mid E \rightsquigarrow A : \text{p}) \end{array}$$



$$\left[\begin{array}{c} A! \{A, B, C, D, E\} \langle \text{h} \rangle \\ \downarrow \\ C! \{A, B, C, D, E\} \langle \text{p} \rangle \end{array} \right]$$

$$e = \{A \text{ 's address}\}_H, \{\text{h}\}_{H^+}, \{\text{hash}(\{A \text{ 's address}\}_H), \text{hash}(\{\text{h}\}_{H^+})\}_{H^-}$$

$$\{\text{margherita}\}_H, \{\text{p}\}_{H^+}, \{\text{hash}(e), \text{hash}(\{\text{margherita}\}_H), \text{hash}(\{\text{p}\}_{H^+})\}_{H^-}$$

Executing projections in Nearbybytes

Executing projections in Nearbybytes

- ▶ Transitions' label univocally mapped to `Nearbybytes` payloads
 - ▶ each output transition invokes `Nearbybytes` API to emit the corresponding payload
 - ▶ each input is a search in the partially-ordered hub

Executing projections in Nearbybytes

- ▶ Transitions' label univocally mapped to `Nearbybytes` payloads
 - ▶ each output transition invokes `Nearbybytes` API to emit the corresponding payload
 - ▶ each input is a search in the partially-ordered hub
- ▶ Easy realisation: generate the automata and interpret them:

Executing projections in Nearbybytes

- ▶ Transitions' label univocally mapped to `Nearbybytes` payloads
 - ▶ each output transition invokes `Nearbybytes` API to emit the corresponding payload
 - ▶ each input is a search in the partially-ordered hub
- ▶ Easy realisation: generate the automata and interpret them:
- ▶ More smartly: "on-demand" generation of the automata leveraging event-notification

Executing projections in Nearbybytes

- ▶ Transitions' label univocally mapped to `Nearbybytes` payloads
 - ▶ each output transition invokes `Nearbybytes` API to emit the corresponding payload
 - ▶ each input is a search in the partially-ordered hub
- ▶ Easy realisation: generate the automata and interpret them:
- ▶ More smartly: "on-demand" generation of the automata leveraging event-notification
- ▶ Compile the automata to actual code (e.g., Typescript or GoLang and use `Nearbybytes` as a library)

Future work

Future work

- ▶ Finalise the implementation

- ▶ Finalise the implementation
- ▶ Study well-formedness conditions of F2F-choreographies for "odd" protocols like

$$A \rightsquigarrow m_1; A \rightsquigarrow m_2; B \rightsquigarrow m_3 + A \rightsquigarrow m_2; B \rightsquigarrow m_4 \quad (1)$$

- ▶ Finalise the implementation
- ▶ Study well-formedness conditions of F2F-choreographies for "odd" protocols like

$$A \rightsquigarrow m_1; A \rightsquigarrow m_2; B \rightsquigarrow m_3 + A \rightsquigarrow m_2; B \rightsquigarrow m_4 \quad (1)$$

- ▶ Consider specific notions of eventual consistency

- ▶ Finalise the implementation
- ▶ Study well-formedness conditions of F2F-choreographies for "odd" protocols like

$$A \rightsquigarrow m_1; A \rightsquigarrow m_2; B \rightsquigarrow m_3 + A \rightsquigarrow m_2; B \rightsquigarrow m_4 \quad (1)$$

- ▶ Consider specific notions of eventual consistency
- ▶ Extend `Nearbytes` with broadcast encryption schemes?

Thank you!

Obscured by clouds (Pink Floyd)

Bridges burning gladly
Merging with the shadows,
Flickering between the lines.
Stolen moments floating softly on the air,
Borne on wings of fire
and climbing higher.

Ancient bonds are breaking,
Moving on and changing sides.
Dreaming of a new day,
Cast aside the other way.

Magic visions stirring,
Kindled by and burning flames rise in her eyes.

The doorway stands ajar,
The walls that once were high.
Beyond the gilded cage,
Beyond the reach of ties.
The moment is at hand.

She breaks the golden band.